Yu-Teng Li (**Kevin**)

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WORK EXPERIENCE

Senior Applied Research Scientist, ***Adobe Research, Foundation Model Team***. **Aug 2023 – Present**

- **Co-led the production run of Firefly Image 5 multimodal pretraining**, a foundation model supporting 4MP text-to-image, instruction editing and layered generation. Designed data curriculum and architecture ablations, debugged training loss instabilities, and drove the final training recipe that runs on 1000+ GPUs with billion-scale datasets.
- **Led multi-reference editing for next-generation video/image generalist Mixture-of-Experts model**. Designed training ablations on editing data recipes and VLM conditioning architectures. Built activation analysis tools that guided improvements in expert routing.
- **Built petabyte-scale video curation for pretraining**, turning raw video into billion-scale multi-shot training clips. Designed a shot detection pipeline tuned per video category by an autoresearch harness against a human-labeled gold set, and transcript-aware multi-shot clip packing that minimizes speech interruptions mid-sentence.
- **Led aesthetic finetuning (SFT) for Firefly Image 4**, delivering the recipe demoed at Adobe MAX. In pretraining, introduced a timestep curriculum that routes synthetic samples to high-noise diffusion steps, broadening concept coverage without inheriting over-saturation.
- **Created UniFusion**, a DiT architecture that replaces separate text and image encoders with multi-layer VLM semantic features, achieving competitive generation & editing and zero-shot multi-image reference generation. **Its layerwise-attention pooling module was later adopted by Krea 2's architecture**.
- Earlier: led personalization for Firefly Image 3; designed inversion-based style transfer with weighted cross-attention, a training-free method for synthesizing paired editing data.

Student Researcher, ***Berkeley Artificial Intelligence Research***. **Feb 2022 – Oct 2023**

- Led research projects on **Active Learning on Image Segmentation** for autonomous driving, in Trevor Darrell's group. [lab website]
- Developed a config-driven experiment package which supports image-, pixel-, and region-based active learning (AL) with multiple sampling heuristics (e.g. entropy, mask autoencoding), eventually leading to the creation of **HALO, the first AL method to surpass the fully supervised baseline** in segmentation benchmarks.

Software Engineering Intern, ***Adobe***. **May 2022 – Aug 2022**

- Developed **Smart Grid for Face Stylization** – a real-time web app (*10 fps*) that stylizes portraits via GAN latent space interpolation. Designed a **layout algorithm** organizing 10k StyleGAN2 embeddings into grids using UMAP and K-means on style affinity scores.
- Designed and trained a CycleGAN-based facial expression translation model to mitigate warping artifacts in Photoshop Liquify Filter.

Student Instructor (Head TA of Discussion), CS182 Deep Learning, ***UC Berkeley***. **Aug 2022 – May 2023**

Student Researcher, ***Vision and Learning Lab @ National Taiwan University***. **Jan 2021 – Oct 2021**

Machine Learning Intern, ***Dell Technologies***. **Jul 2021 – Aug 2021**

SELECTED PUBLICATIONS

UniFusion: VLM as Unified Encoder in Image Generation and Editing [ArXiv] [Website] – **ICLR 2026 Multimodal Intelligence Workshop**. **Architecture adopted by Krea 2.** Challenges the premise that semantic embeddings are insufficient for pixel-level generation: conditioning a DiT solely on multi-layer VLM features without VAE signals, surpasses Flux and BAGEL on DPG-Bench and rivals Qwen-Image-Edit on editing. The unified multimodal space further yields emergent capabilities such as zero-shot multi-ref generation and cross-task knowledge transfer.

Squeezing Capacity from Multimodal LLMs for Subject-driven Generation – **NeurIPS 2026**. Achieves better subject identity preservation and harmonization than prior methods by aggregating multi-layer VLM features and a multi-stage denoising strategy to balance the details in VAE and VLM cross-modal semantics.

Towards Text-Guided Attribute-Disentangled Multimodal Representation Learning – **CVPR 2026**. Formulates Queryable Attribute Representation Extraction (QARE) and introduces the first benchmark for attribute-disentangled multimodal embeddings. Proposes a training-free method that substantially outperforms SigLIP and contrastively trained methods such as VLM2Vec on attribute-conditioned retrieval.

Hyperbolic Active Learning for Semantic Segmentation under Domain Shift [ArXiv] – **ICML 2024**. HALO uses a hyperbolic neural network for pixel-level active learning in semantic segmentation, and is the first active learning approach that surpasses the performance of supervised domain adaptation with only 1% labeled data on synthetic-to-real benchmarks such as GTAV → Cityscapes.

EDUCATION

University of California, Berkeley **Aug 2019 – May 2023**

B.S. Electrical Engineering & Computer Science (GPA: 3.7 / 4)